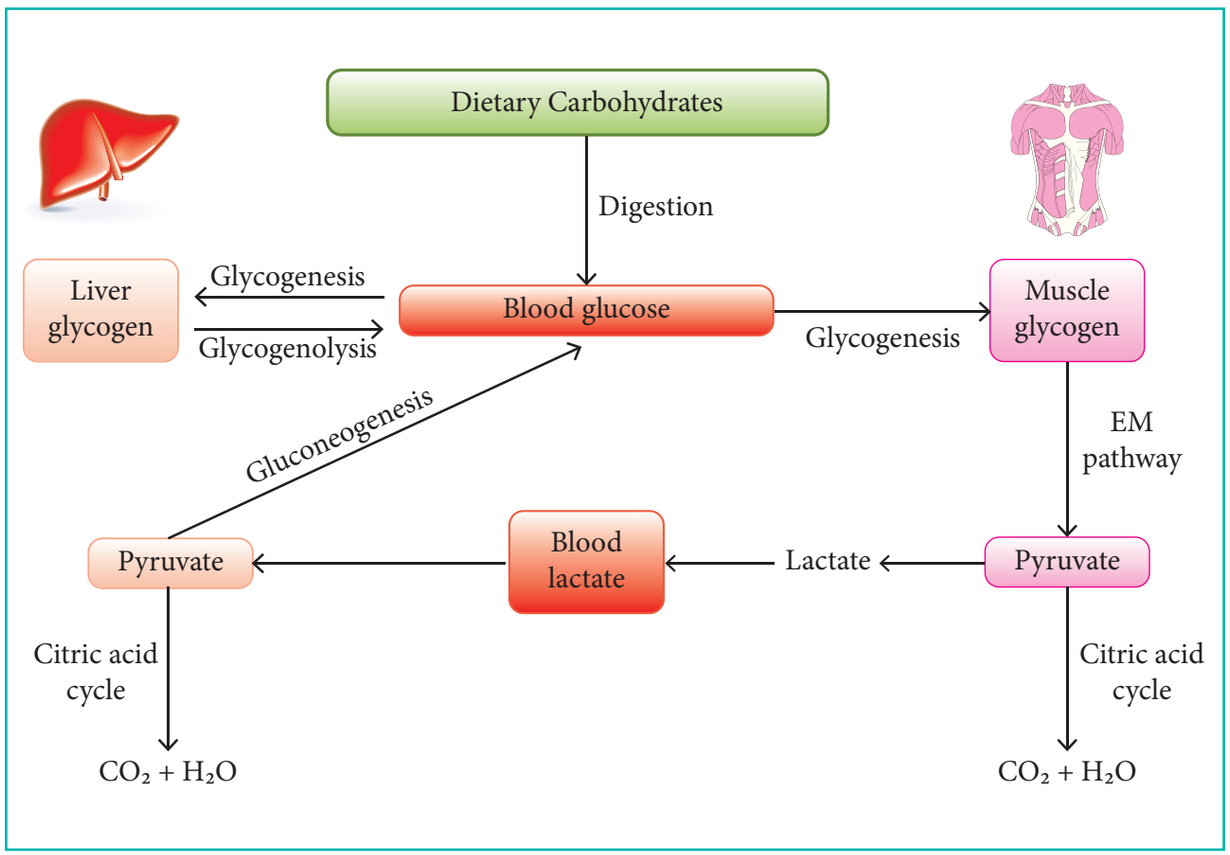
3.2 CARBOHYDRATE AS A SOURCE OF ENERGY

The major function of carbohydrate is to serve as a fuel. The catabolic enzymatic oxidation of glucose provides energy and the metabolic intermediates are used in various biosynthetic reactions. We require energy to perform any work such as playing. Reading, etc., Carbohydrate metabolism provides us the required energy. Glucose is oxidized via EMP pathway (Embden - Meyerhodf - Parnas) also known as glycolysis. During glycolysis, glucose is converted into pyruvate. Pyruvate is then converted into acetyl CoA which is fully oxidised to carbon dioxide via the citric acid cycle in the presence of oxygen (aerobic conditions). Under oxygen limiting conditions the pyruvate is converted into lactate. Lactate is then transported to liver where it is metabolised further. Glucose can be synthesised from pyruvate by gluconeogenesis pathway which will be discussed later in detail.



The liver maintains the glucose level in the blood. When blood glucose is in excess, it is converted to glycogen in the liver and muscle tissue. The synthesis of glycogen form glucose is called **glycogenesis**. In addition, glycolytic pathway is also accelerated.

When the glucose level is low, the glycogen is hydrolysed to produce glucose. The hydrolysis of glycogen to glucose is called **glycogenolysis**. The glucose level is also increased by activating the gluconeogenesis pathway.

XII U3 Carbohydrates.indd 65 4/8/2020 3:06:51 PM